

HOMEWORK 6

Track 2

Due date: November 16, 2023.

Reading assignment: Revise chapters 1, 2 and 3 of the class lecture notes. Read chapter 5 of the class lecture notes. Read the notes on difference equations and differential equations posted on the class website. Read chapters 6 & 7 of the text book. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

Exercise 1. Prove the following statement of the ***CS decomposition*** (thin version). Consider the matrix

$$Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}, \quad Q_1 \in \mathbb{R}^{m_1 \times n}, Q_2 \in \mathbb{R}^{m_2 \times n},$$

where $m_1 \geq n$ and $m_2 \geq n$, and Q has orthonormal columns. Show that there exists orthogonal matrices U_1 , U_2 and V , such that

$$\begin{pmatrix} U_1 & 0 \\ 0 & U_2 \end{pmatrix}^T \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} V = \begin{pmatrix} \begin{pmatrix} C \\ 0 \end{pmatrix} \\ \begin{pmatrix} 0 \\ S \end{pmatrix} \end{pmatrix},$$

where

$$C = \begin{pmatrix} \cos \theta_1 & & & \\ & \cos \theta_2 & & \\ & & \ddots & \\ & & & \cos \theta_n \end{pmatrix}, \quad S = \begin{pmatrix} \sin \theta_1 & & & \\ & \sin \theta_2 & & \\ & & \ddots & \\ & & & \sin \theta_n \end{pmatrix},$$

and

$$0 \leq \theta_1 \leq \theta_2 \leq \dots \leq \theta_n \leq \pi/2.$$

Hint: SVD of Q_1 , followed by QR factorization of $Q_2 V$ (with no column pivoting).

Exercise 2. Let λ be a non-zero complex number. Consider the equations

$$\begin{aligned} u[n] &= \lambda u[n-1] + f[n], \\ \sum_{n=-\infty}^{\infty} b[n] u[n] &= \varphi, \end{aligned} \quad n \in \mathbb{Z},$$

where $f[n]$ and $b[n]$ are assumed to be known sequences of complex numbers, φ is a known complex number, and $\mathbb{Z}$ denotes the set of all integers. In the following problems make any necessary (mild) assumptions on the sequence $b[n]$.

- Show that there is at most one sequence $u[n]$ that satisfies the above set of equations.
- Find $u[n]$ when $f[n] = z^n$, for some complex number z .
- Show that $u[n]$ always exists.
- Extra credit:* What are the equations for the left null-space associated with the above set of linear equations? What is the dimensionality of the left null-space? It is not 0! Why is this surprising? (*Warning:* There are some subtle issues here.)

Hint: Look at my notes on difference equations to see how to compute $u[n]$.

Exercise 3. Let λ be a non-zero complex number. Consider the equations

$$\begin{aligned} \frac{d}{dt} u(t) &= \lambda u(t) + f(t), \\ \int_{-\infty}^{\infty} b(t) u(t) dt &= \varphi, \end{aligned} \quad t \in \mathbb{R},$$

where $f(t)$ and $b(t)$ are assumed to be known complex-valued functions on $\mathbb{R}$, φ is a known complex number, and $\mathbb{R}$ denotes the set of all real numbers. In the following problems make any necessary (mild) assumptions on the functions $f(t)$, $b(t)$ and the putative solutions $u(t)$. (*Warning:* Trying to find the least restrictive assumptions can become very tricky and convoluted, and is not needed.)

- a) Show that there is at most one function $u(t)$ that satisfies the above set of equations.
- b) Find $u(t)$ when $f(t) = e^{st}$, for some complex number s .
- c) Show that $u(t)$ always exists.
- d) *Extra credit:* What are the equations for the left null-space associated with the above set of linear equations? What is the dimensionality of the left null-space? It is not 0! (*Warning:* Compared to the previous exercise there are even subtler issues here. I would recommend ignoring them in the first pass.)

Hint: Look at my notes on differential equations to see how to compute $u(t)$.